- Muhammad Haris Ahsan

Week 05 – Lab Assignment

Objective: To demonstrate proficiency in manipulating Python collection types using built-in methods.

1. Working with List Methods

Lists are ordered, mutable collections used to store multiple items in a single variable.

Method	Description
append()	Adds an element to the end of the list.
insert()	Adds an element at a specific index.
remove()	Removes the first occurrence of a specific value.
sort()	Organizes the list in ascending/alphabetical order.
reverse()	Flips the order of the elements.

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Example code:

```python
# Part 1: List Operations

print("===== Working with LIST Methods =====")

# Initialization
fruits = ["apple", "banana", "cherry"]
print(f"Original List: {fruits}")

# Adding items
fruits.append("mango")
print(f"After append('mango'): {fruits}")

fruits.insert(index: 1, object: "orange")
print(f"After insert(1, 'orange'): {fruits}")

# Removing items
fruits.remove("banana")
print(f"After remove('banana'): {fruits}")

# Organizing
fruits.sort()
print(f"After sort() (Alphabetical): {fruits}")

fruits.reverse()
print(f"After reverse(): {fruits}")

# Information & Access
print(f"Final List Length: {len(fruits)}")
print(f"First element (Index 0): {fruits[0]}")
```

Output:

```
C:\Python314\python.exe "D:\dev\Py - PR\Labs\Lab Assignments\main.py"
===== Working with LIST Methods =====
Original List: ['apple', 'banana', 'cherry']
After append('mango'): ['apple', 'banana', 'cherry', 'mango']
After insert(1, 'orange'): ['apple', 'orange', 'banana', 'cherry', 'mango']
After remove('banana'): ['apple', 'orange', 'cherry', 'mango']
After sort() (Alphabetical): ['apple', 'cherry', 'mango', 'orange']
After reverse(): ['orange', 'mango', 'cherry', 'apple']
Final List Length: 4
First element (Index 0): orange

Process finished with exit code 0
```

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2. Working with Dictionary Methods

Dictionaries store data in **key:value** pairs, allowing for fast lookups and organized data storage.

Method	**Description**
get()	Safely retrieves a value using its key.
update()	Modifies an existing key or adds a new one.
pop()	Removes a specific key and returns its value.
keys() / values()	Returns a view object of all keys or all values.
clear()	Removes all items from the dictionary.

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Example code:

```python
# Part 2: Dictionary Operations

print("\n===== Working with DICTIONARY Methods =====")

# Initialization
student = {"name": "Ali", "age": 20, "grade": "A"}
print(f"Original Dictionary: {student}")

# Accessing Data
print(f"Retrieved Name: {student.get('name')}")

# Modifying Data
student["city"] = "Lahore"
print(f"After adding 'city': {student}")

student.update({"grade": "A+"})
print(f"After update('grade'): {student}")

# Deleting Data
student.pop("age")
print(f"After pop('age'): {student}")

# View Objects
print(f"Dictionary Keys: {list(student.keys())}")
print(f"Dictionary Values: {list(student.values())}")

# Resetting
student.clear()
print(f"After clear(): {student}")
```

Output:

```
C:\Python314\python.exe "D:\dev\Py - PR\Labs\Lab Assignments\main.py"

===== Working with DICTIONARY Methods =====
Original Dictionary: {'name': 'Ali', 'age': 20, 'grade': 'A'}
Retrieved Name: Ali
After adding 'city': {'name': 'Ali', 'age': 20, 'grade': 'A', 'city': 'Lahore'}
After update('grade'): {'name': 'Ali', 'age': 20, 'grade': 'A+', 'city': 'Lahore'}
After pop('age'): {'name': 'Ali', 'grade': 'A+', 'city': 'Lahore'}
Dictionary Keys: ['name', 'grade', 'city']
Dictionary Values: ['Ali', 'A+', 'Lahore']
After clear(): {}

Process finished with exit code 0
```